

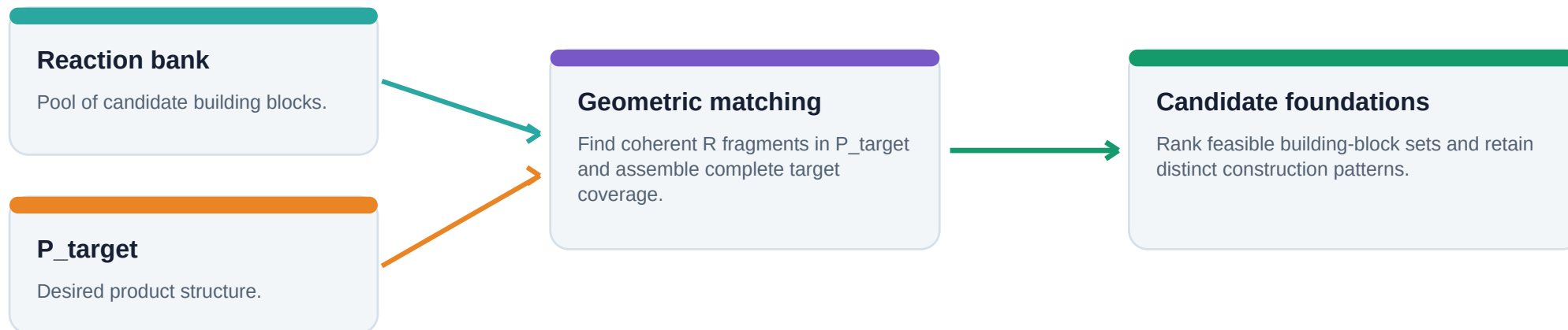
Purpose: Find Building Blocks for P_target

Use a reaction-bank pool to identify structural foundations for retrosynthesis

Purpose

Given P_target, find feasible building blocks whose conserved molecular pieces can serve as the foundation for retrosynthesis.

General flow



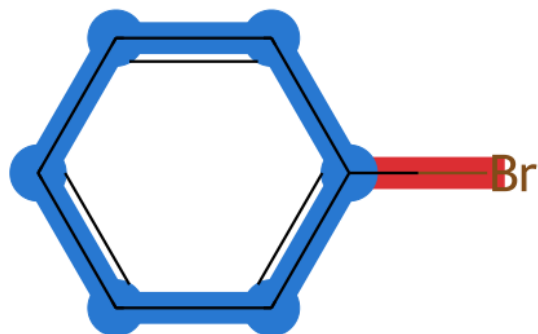
CURRENT SCOPE: PURE MOLECULAR GEOMETRY

The workflow identifies structurally compatible building blocks. It does not yet prove chemical reactivity or construct a complete multi-step synthesis route.

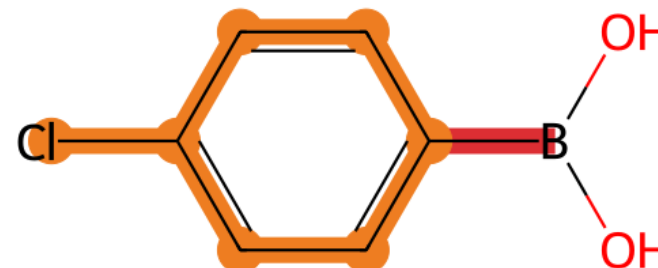
Example 1: two building blocks cover P_target

Complete 2D molecules are shown; identical colors are assigned by the computed atom mapping

R1: bromobenzene

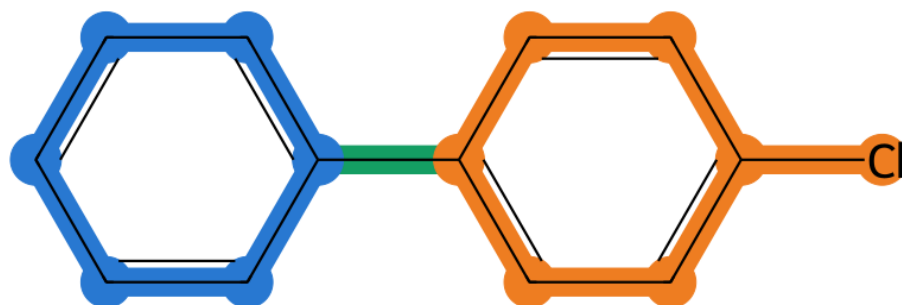


R2: 4-chlorophenylboronic acid



+

P_target: 4-chlorobiphenyl



What the mapping shows

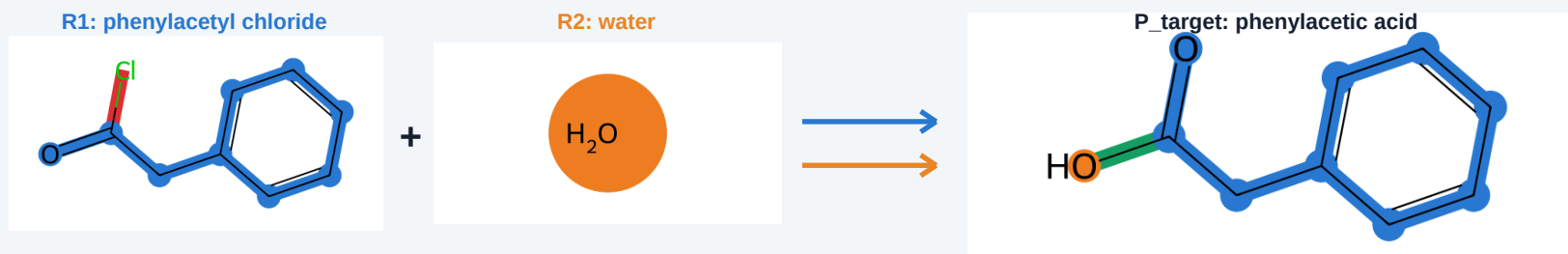
- blue atoms come from R1
- orange atoms come from R2
- red source groups are not retained
- green is the new bond joining the pieces

Example 2: alternative building-block patterns

One best representative is shown for each distinct structural division of phenylacetic acid

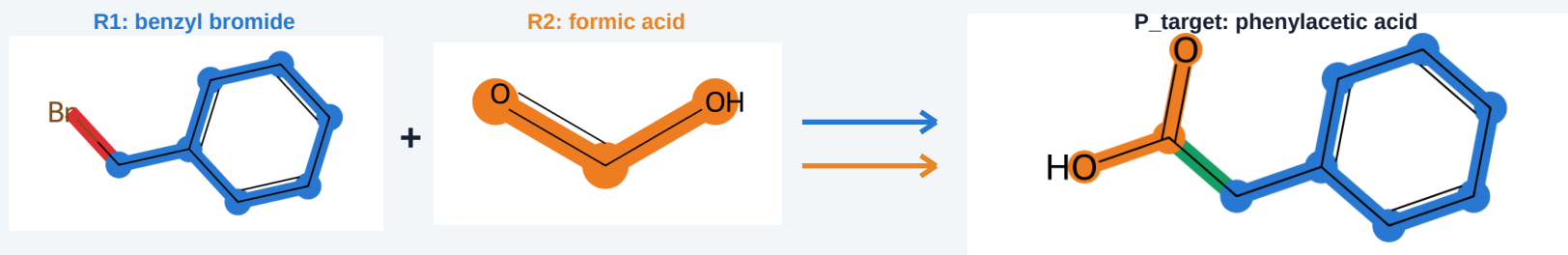
Pattern A

carbonyl framework retained



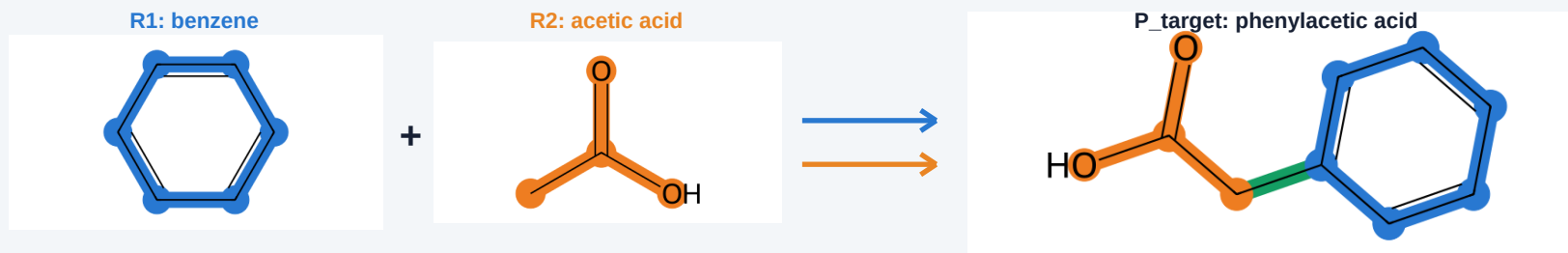
Pattern B

benzyl and carboxyl modules



Pattern C

aryl and acetic-acid modules



- Every row completely covers the same P_target, but divides it between R1 and R2 differently.
- The workflow returns one representative per pattern; chemical feasibility is reviewed separately.

Example 3: one building block contributes multiple pieces

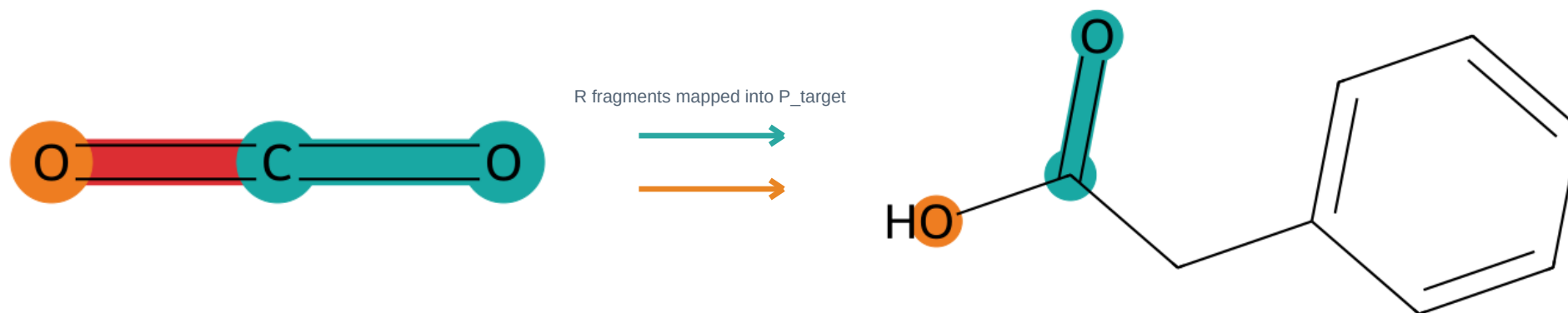
CO₂ is recognized within the carboxyl group of phenylacetic acid

R: carbon dioxide

The carbonyl piece is recognized first. The second oxygen is then retained as another useful piece of the same source.

P_target: phenylacetic acid

Only the colored carboxyl atoms are owned by this candidate. The aromatic side of P remains available to other sources.



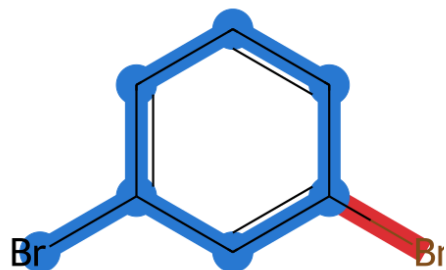
What the colors show

- The cyan carbonyl is the main connected match.
- The orange oxygen is recognized as a second useful piece from the same R.
- Together, the colored atoms account for the CO₂ contribution; the uncolored target remains available to other building blocks.

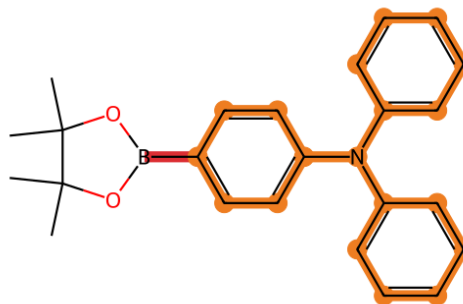
Example 4: the same workflow on a larger target

R1 and R2 are mapped into a complete P_target using the same color convention

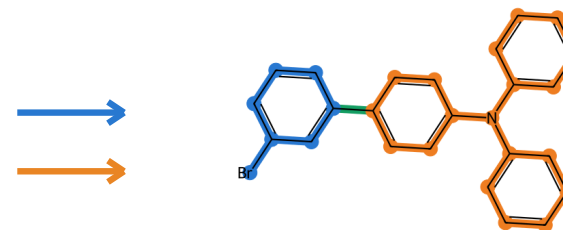
R1: 1,3-dibromobenzene



R2: triarylamine BPin



P_target: assembled product



● retained smaller aryl module

● retained triarylamine module

● new inter-module bond

What this example demonstrates

- The large triarylamine system remains one coherent orange contribution from R2.
- The smaller blue aryl module comes from R1.
- Unused bromine and the BPin unit are uncolored because they are not retained in P_target.
- Together, the blue and orange regions cover the complete target; green marks their new bond.

Harder Example: Two Steps with Multiple R

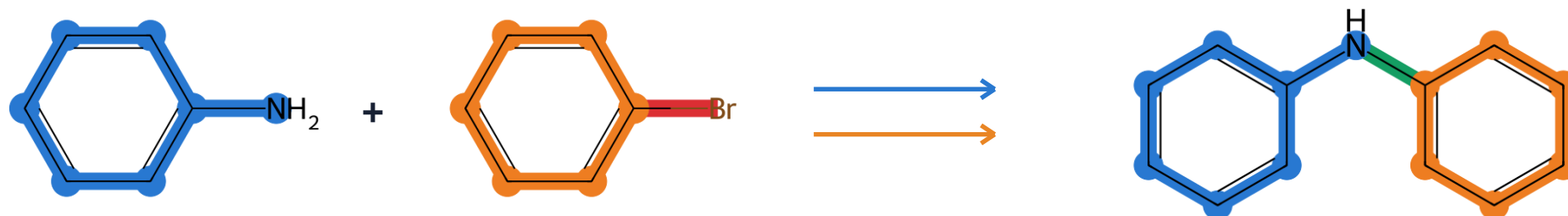
The same geometric workflow is applied recursively; bromobenzene is used twice

Step 1: construct the intermediate

R1: aniline

R2: bromobenzene, copy 1

P1: diphenylamine

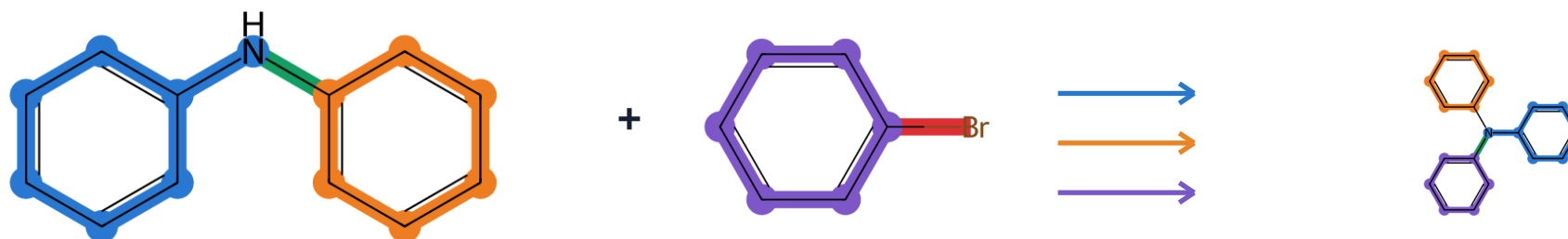


Step 2: expand the intermediate to P_target

R1: P1 intermediate

R3: bromobenzene, copy 2

P_target: triphenylamine

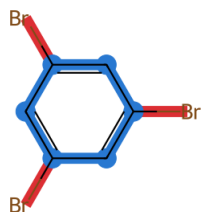


- Both stages independently achieve complete explicit-H target coverage without a branch-cap hit.
- The direct foundation search also recovers aniline plus two copies of bromobenzene; a route planner is needed to introduce P1 as the intermediate.

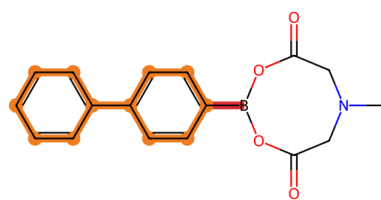
Really Difficult Example: Seven-Ring Star Target

One central building block plus three repeated biphenyl-arm building blocks

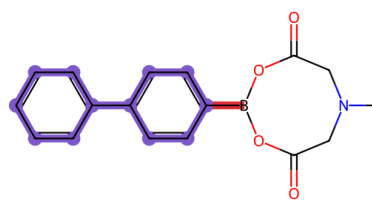
R1: tribromobenzene core



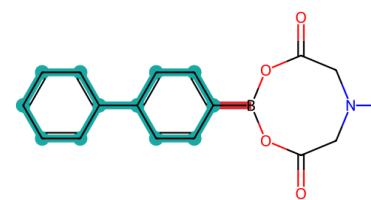
R2: biphenyl arm, copy 1



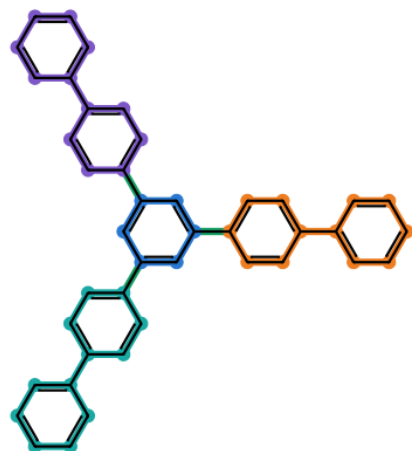
R3: biphenyl arm, copy 2



R4: biphenyl arm, copy 3



P_target: complete seven-ring product



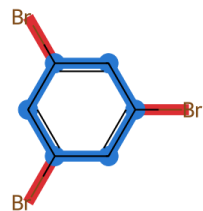
Computed result

- 42 heavy atoms and 72 explicit atoms are fully covered.
- Four R copies are selected: one core and three identical arm structures.
- Three inter-module bonds are formed; all searches finish below branch cap 100.
- This gives the foundations, but does not choose the order or chemistry of the three coupling stages.

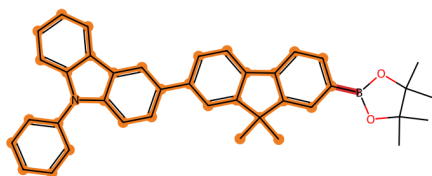
Extreme Size Test: 183 Explicit Atoms

A 108-heavy-atom target assembled from one core and three repeated OLED building blocks

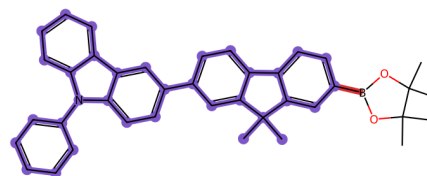
R1: central core



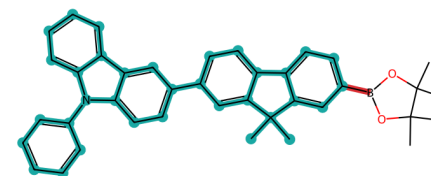
R2: large arm, copy 1



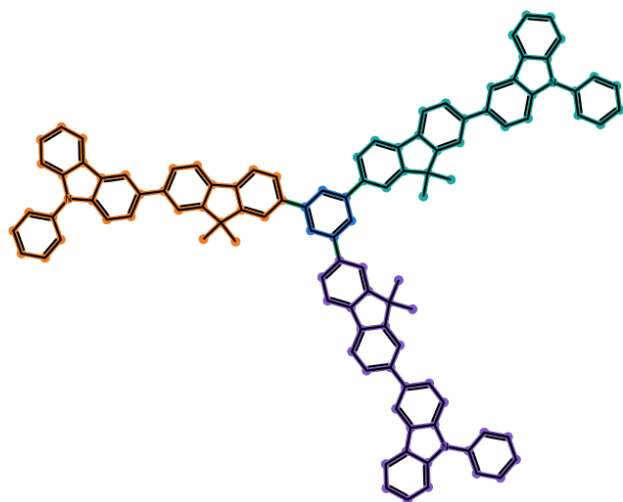
R3: large arm, copy 2



R4: large arm, copy 3



P_target: complete 108-heavy-atom product



Saved computed result

- Complete coverage: 9 + 58 + 58 + 58 explicit atoms.
- Three product bonds join the four source-derived modules.
- Detection takes 147.6 seconds; final assembly takes 0.004 seconds.
- Maximum branch count is four, with no cap hit: source scanning, not branching, is the bottleneck.

Open Questions: From Geometry to Retrosynthesis

The geometric candidates are a foundation; two capabilities are still required

CURRENT OUTPUT

Structurally compatible building-block sets that cover P_target from a pure geometrical perspective.

1. Chemical reactivity rules

Can the proposed transformation actually occur?

- Which bonds can be formed or broken by known chemistry?
- Are functional groups, reagents, catalysts, and conditions compatible?
- Will chemo-, regio-, and stereoselectivity be acceptable?

2. Multi-step route construction

How do the foundations become a complete route?

- Use accepted building blocks or intermediates as new targets.
- Connect validated single steps into a retrosynthesis path.
- Control branching and rank complete routes by chemical feasibility.

